Junfeng Zou

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Research Interests

My research focuses on robot planning and control, particularly the integration of model-based control and embodied learning for robust and reliable robotic manipulation, locomotion, and autonomous systems.

Education

The Chinese University of Hong Kong, Shenzhen Shenzhen, China
M.Phil. in Computer and Information Engineering Sep 2025 – Expected Jun 2027
Supervisor: Prof. Fangxun Zhong **Research Group:** Intelligent Healthcare & Autonomous Robotics Lab

Northwestern Polytechnical University Xi'an, China
B.E. in Robotics Sep 2021 – Jun 2025
GPA: 3.69/4.10
Thesis: *Simulation and Analysis of Gait Planning Algorithms for Open-Source Humanoid Robots*

Research Experience

Vision-Guided Robotic Nasal Endoscope Control and Safety Supervision System Jul 2025 – Present
Research Project, Intelligent Healthcare & Autonomous Robotics Lab, CUHK-Shenzhen
This project focuses on robot-assisted endoscope control for nasal surgery, using a robotic system to help surgeons maintain a stable and safe surgical view during long procedures. The project was conducted in collaboration with an experienced otolaryngologist.

- Developed a visual servoing system for robotic nasal endoscopy with Dobot CR5, integrating TCP/IP communication, ServoJ control, camera calibration, and real-time feature tracking.
- Designed and validated a hierarchical MPC controller for IBVS under RCM constraints, maintaining the RCM deviation below 1.6 mm during real-robot experiments.
- Integrated joint and velocity constraints, Kalman filtering, and fault-tolerant feature handling; built endoscopic vision modules for instrument segmentation, distance-state estimation, and directional collision-risk assessment.

EtherCAT-Based Motor Control System for da Vinci Surgical Instruments Dec 2025 – Jun 2026
Engineering Project, Intelligent Healthcare & Autonomous Robotics Lab, CUHK-Shenzhen
This project focused on low-level actuation control for a da Vinci surgical instrument kit, using EtherCAT to control Denali XCR drive boards and MPK motors as a real-time hardware interface for surgical tool motion control.

- Built a ROS 2, EtherCAT, and `ros2_control`-based multi-axis motor control framework for Denali XCR drives and MPK motors.
- Implemented CiA 402 drive control, including fault reset, drive enabling, operation-mode switching, and position/velocity command execution.
- Developed a motor debugging and monitoring interface for real-hardware testing, parameter tuning, and diagnosis.

Deep Reinforcement Learning-Based Locomotion Control for a Humanoid Robot Dec 2024 – Jun 2025
Undergraduate Thesis, Northwestern Polytechnical University
This project focused on developing PPO-based locomotion policies for the open-source XBot-L humanoid robot, enabling stable walking across flat and complex terrains.

- Built a GPU-parallel reinforcement learning environment in Isaac Gym and implemented an Actor–Critic PPO training pipeline for 12-DoF leg control.
- Designed multi-objective reward functions for velocity tracking, posture stability, gait phase regulation, and joint motion, incorporating privileged observations, domain randomization, action noise, and random disturbances.
- Trained and compared locomotion policies on flat, rough, sloped, and stair terrains, evaluating their performance through base trajectories, velocity tracking, and walking stability.

Humanoid Robot Assembly and EtherCAT-Based Joint Actuation Integration Jul 2024 – Jan 2025
Visiting Research Project, Adaptive Robotic Controls Lab (ArcLab), The University of Hong Kong
This project focused on the assembly and low-level integration of an open-source humanoid robot, with primary responsibility for EtherCAT-based joint motor control and motor wiring integration.

- Participated in the mechanical assembly and electrical integration of the robot, including joint actuator installation, motor wiring, and system commissioning.
- Developed and tested an EtherCAT-based multi-joint motor control system, covering slave configuration, drive enabling, cyclic communication, and synchronized motor operation.

Selected Projects

LoRA-Fine-Tuned OpenVLA for Robotic Drink Pouring Jan 2026 – May 2026

Final Project, Course: Robotic Manipulation, CUHK-Shenzhen

This project explored the training and deployment workflow of VLA models on a real-world robotic system, providing hands-on experience with VLA fine-tuning and closed-loop robotic inference.

- Built a real-world robotic drink-pouring VLA system based on OpenVLA-7B and LoRA.
- Designed a vision-language-action supervision pipeline for predicting 7-DoF end-effector delta actions from language instructions and visual observations.
- Deployed the VLA inference pipeline in a real physical setup, denormalizing predicted 7-D actions and mapping them to robot arm and gripper control for closed-loop execution from visual observations.

Multi-Finger Dexterous Hand Assembly and Interactive Control System Jan 2026 – May 2026

Final Project, Course: Robotics and Intelligent Systems, CUHK-Shenzhen

Based on the open-source DexHand hardware design, this project involved building a multi-finger dexterous hand, including 3D printing, mechanical assembly, servo wiring, control-board integration, motion-range calibration, and software development for an interactive rock-paper-scissors game.

- Built the mechanical and electronic hardware system of a multi-finger dexterous hand, including servo installation, joint tuning, motion-range calibration, and basic pose testing.
- Developed a control interface based on RP2350, serial communication, and Wi-Fi, converting finger-pose parameters into servo-angle commands for execution.
- Implemented real-time gesture recognition with MediaPipe and OpenCV to detect rock, paper, and scissors gestures and control DexHand to perform corresponding game poses.

Publications

The following publications resulted from my undergraduate research in flexible sensing.

[J1] C. Dong, Y. Bai, **J. Zou**, J. Cheng, Y. An, Z. Zhang, Z. Li, S. Lin, S. Zhao, and N. Li, "Flexible capacitive pressure sensor: material, structure, fabrication and application."
Nondestructive Testing and Evaluation, 2024.

[C1] Y. An, **J. Zou**, Y. Bai, C. Dong, J. Cheng, and N. Li, "Design and Testing of Flexible Pressure and Temperature Sensing Capacitive Sensors."
2024 IEEE International Instrumentation and Measurement Technology Conference (I2MTC), 2024.

Awards and Honors

- First Prize, Swarm Robot Task Competition, National Robot Championship 2022
- First Prize, Semi-Autonomous Robot Soccer Competition, National Robot Championship 2022
- Second Prize, China Marine Vehicle Design and Construction Contest 2023
- Second Prize, HUAWEI CUP National Undergraduate IoT Design Contest 2022

Technical Skills

Programming: Python, C/C++, MATLAB, Linux
Robotics and Control: ROS 2, EtherCAT, MPC, visual servoing, robot manipulation
Learning and Vision: PyTorch, Isaac Gym/Sim, reinforcement learning, OpenCV, YOLO

Teaching

Teaching Assistant, The Chinese University of Hong Kong, Shenzhen Spring 2026

CSC1006: Artificial Intelligence for Science and Engineering

Conducted tutorial sessions, answered students' questions, proctored examinations, and graded exam papers.